ECEN-662: Homework 1

Due on January 27, 2026 at 11:59 PM

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Problem 1

Let $X \sim p_\theta(x)$, where

$$p_\theta(x) = \left(\frac{\theta}{2}\right)^{|x|} (1-\theta)^{1-|x|}, \quad x \in \{-1, 0, 1\} \quad \text{and} \quad \theta \in (0,1)$$

i) Does $p_\theta(x)$ belong to the exponential family?

ii) Is $|X|$ a complete sufficient statistic?

iii) Is X complete?

Solution

Part (i):

Part (ii):

Part (iii):

Problem 2

Let $X = (X_1, \ldots, X_n)$, where X_i's are i.i.d. $\mathcal{N}(\theta, \theta)$, and $\theta > 0$ is an unknown parameter. Determine a complete sufficient statistic for θ and carefully justify your answer.

Solution

Problem 3

Let $X = (X_1, \ldots, X_n)$, where X_i's are i.i.d. $\mathcal{N}(\theta, \theta^2)$, and $\theta > 0$ is an unknown parameter.

i) For $n = 1$, show that $T = X$ is a minimum sufficient statistic but is not a complete sufficient statistic.

ii) Show that the sufficient statistic

$$T = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right)$$

is not complete for any $n \geq 2$.

Solution

Problem 4

Prove the following law of total variance:

$$\mathrm{Var}[Y] = \mathbb{E}[\mathrm{Var}[Y|Z]] + \mathrm{Var}[\mathbb{E}[Y|Z]]$$

for any real-valued random variable Y.

Solution